Manuscript for Parts 1–5

Modeling, Hamiltonian Structure, CHNN Construction, and $\mathcal{K}$-Symplectic Algorithms

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1. Flexible-Rod PDE

This part is the starting point of the whole project. The objective is to derive a unified dynamical PDE for a flexible rod in microgravity by combining four classical ingredients: Kirchhoff/Cosserat rod kinematics, balance laws, constitutive relations, and four external-force models. The resulting equation is the variable-coefficient nonlinear wave equation (9). In the subsequent parts it is discretized by a Galerkin procedure, rewritten in Hamiltonian form, generalized to the Birkhoff framework, and eventually connected to the training of a complex-valued Hamiltonian neural network.

1.1 Rod Geometry and Kinematics

Kirchhoff theory models the rod as a spatial curve that can bend, twist, and stretch. An orthonormal triad is attached to each cross-section. The basic variables are listed in Table 1.1.

Table 1.1: Basic variables for rod geometry and kinematics.

Symbol	Meaning	Unit
$\boldsymbol{r}(s,t)$	centerline position vector of the rod	m
s	Lagrangian arc-length coordinate along the rod axis in the reference configuration	m
t	time	s
$\boldsymbol{t}$	unit tangent vector after deformation; taken as $\boldsymbol{d}_3$	dimensionless
v	stretch ratio, $v = 1 + \varepsilon$	dimensionless
ε	axial stretch strain	dimensionless
$\boldsymbol{u}$	Darboux vector describing local bending and twisting rates	rad/m
$\{\boldsymbol{d}_1, \boldsymbol{d}_2, \boldsymbol{d}_3\}$	orthonormal director frame attached to the cross-section	dimensionless

1.1.1 Tangent relation

The kinematic tangent relation is

$$\frac{\partial \boldsymbol{r}}{\partial s} = v\,\boldsymbol{t}, \qquad v = \left\| \frac{\partial \boldsymbol{r}}{\partial s} \right\|. \tag{v}$$

Its meaning is straightforward: the derivative of the position vector with respect to arc length determines both the tangent direction and the local stretching. Under the inextensible Kirchhoff assumption one has $v = 1$ and therefore $\varepsilon = 0$. In the general extensible case $v = 1 + \varepsilon$ and $\boldsymbol{t}$ is identified with $\boldsymbol{d}_3$.

1.1.2 Darboux-vector relation

The evolution of the director frame along the rod axis is governed by

$$\frac{\partial \boldsymbol{d}_i}{\partial s} = \boldsymbol{u} \times \boldsymbol{d}_i. \tag{vi}$$

The vector $\boldsymbol{u}$ is the angular-rate vector of the local frame with respect to the axial coordinate. Moving from s to $s + ds$, the frame rotates by an angle $|\boldsymbol{u}|ds$ about the direction of $\boldsymbol{u}$. Its components correspond to two bending curvatures and one twist rate.

1.1.3 Curvature vector

A convenient curvature representation is

$$\boldsymbol{k} \approx \frac{\partial \boldsymbol{t}}{\partial s} \times \boldsymbol{t} = \frac{1}{v}\frac{\partial}{\partial s}\left(\frac{1}{v}\frac{\partial \boldsymbol{r}}{\partial s}\right) \times \frac{\partial \boldsymbol{r}}{\partial s}.$$

The magnitude of $\boldsymbol{k}$ is the curvature κ, and its direction points toward the center of curvature.

1.2 Balance Laws and Constitutive Relations

The geometric relations describe how the rod deforms. The balance and constitutive equations explain why the deformation occurs.

1.2.1 Translational balance

$$\rho A \frac{\partial^2 \boldsymbol{r}}{\partial t^2} = \frac{\partial \boldsymbol{n}}{\partial s} + \boldsymbol{f}_{\text{ext}}. \tag{i}$$

Here ρ is the material density, A is the cross-sectional area, $\boldsymbol{n}$ is the internal force vector, and $\boldsymbol{f}_{\text{ext}}$ is the external force per unit length. The equation is Newton's second law applied to an infinitesimal rod element. The external load is decomposed as

$$\boldsymbol{f}_{\text{ext}} = \boldsymbol{f}_{\text{mag}} + \boldsymbol{f}_{\text{hydro}} + \boldsymbol{f}_{\text{cont}} + \boldsymbol{f}_{\text{gravity}}.$$

In the microgravity setting one takes $\boldsymbol{f}_{\text{gravity}} \approx 0$, so the practical forcing consists of magnetic, hydrodynamic, and contact contributions.

1.2.2 Rotational balance

$$\frac{\partial \boldsymbol{m}}{\partial s} + \frac{\partial \boldsymbol{r}}{\partial s} \times \boldsymbol{n} + \boldsymbol{l}_{\text{mag}} + \boldsymbol{l}_{\text{cont}} = J\frac{\partial \boldsymbol{w}}{\partial t} + \boldsymbol{w} \times (J\boldsymbol{w}). \tag{ii}$$

The vector $\boldsymbol{m}$ is the internal moment, $\boldsymbol{w}$ is the cross-sectional angular velocity, and $J = \rho I$ is the rotational inertia per unit length. The nonlinear term $\boldsymbol{w} \times (J\boldsymbol{w})$ is the gyroscopic/precessional effect and becomes important under large deformation or fast rotation.

1.2.3 Force–strain constitutive law

$$\boldsymbol{n} = K_s(\boldsymbol{\Gamma} - \boldsymbol{\Gamma}_0). \tag{iii}$$

The vector $\boldsymbol{\Gamma}$ is the shear/extension strain, $\boldsymbol{\Gamma}_0$ is the strain in the natural stress-free state, and K_s is the shear/extension stiffness matrix. The meaning is linear elasticity: the internal force is proportional to the deviation of strain from its natural state. To avoid ambiguity with the position vector, the shear/extension strain is denoted here by the standard Cosserat symbol $\boldsymbol{\Gamma}$.

1.2.4 Moment–curvature constitutive law

$$\boldsymbol{m} = K_b(\boldsymbol{u} - \boldsymbol{u}_0), \qquad K_b = \mathrm{diag}(EI_1, EI_2, EI_3). \tag{iv}$$

The curvature/twist vector is $\boldsymbol{u} = (k_1, k_2, \tau)^T$, $\boldsymbol{u}_0$ is the natural curvature, and E is Young's modulus. The diagonal form implies uncoupled bending in the two principal directions and torsion. A technical remark is in order: strictly speaking, the torsional stiffness is usually written as GJ rather than EI_3, so the third diagonal entry should be interpreted with care.

1.3 External-Force Models

The term "four-field coupling" refers to the combination of magnetic, fluid, contact, and gravity effects.

1.3.1 Magnetic force

$$\boldsymbol{f}_{\mathrm{mag}} = \oint \boldsymbol{\sigma}_m \cdot \boldsymbol{n}\, dl. \tag{4}$$

The integral accumulates the magnetic stress on the boundary surrounding the rod cross-section. This is advantageous because one does not need to resolve the internal magnetization distribution explicitly.

1.3.2 Hydrodynamic force

$$\boldsymbol{f}_{\mathrm{hydro}} = -\rho_f A C_m \left(\frac{\partial^2 \boldsymbol{r}}{\partial t^2} - \dot{\boldsymbol{V}}_f \right) - \frac{1}{2}\rho_f C_d \, |\boldsymbol{V}_r| \, \boldsymbol{V}_r. \tag{5}$$

Here ρ_f is the fluid density, C_m the added-mass coefficient, C_d the drag coefficient, $\boldsymbol{V}_f$ the fluid velocity, and $\boldsymbol{V}_r = \partial_t \boldsymbol{r} - \boldsymbol{V}_f$ the relative velocity. The first term is an inertia/additional-mass contribution; the second is a quadratic drag force opposing the relative motion. A dimensional-consistency remark is important here: the drag term should contain the outer diameter D of the rod in order to yield a force density in N/m.

1.3.3 Contact force

$$\boldsymbol{f}_{\mathrm{cont}} = \begin{cases} -k_p (d - d_0)^n \, \boldsymbol{n}_{\mathrm{wall}} - \mu_f |F_{\mathrm{normal}}| \, \boldsymbol{t}_{\mathrm{fric}}, & d \le d_0, \\ \boldsymbol{0}, & d > d_0. \end{cases} \tag{6}$$

The normal term is a Hertz-type nonlinear repulsion, and the tangential term is Coulomb friction. When $d < d_0$ and $n = 1.5$, the literal expression $(d - d_0)^n$ would be negative at the base; physically the intended magnitude is $(d_0 - d)^n$ with direction controlled by $-\boldsymbol{n}_{\mathrm{wall}}$.

1.3.4 Microgravity term

The gravity term is kept symbolically as $\boldsymbol{f}_{\text{gravity}}$ to preserve generality, but in the target application it is effectively zero.

1.4 Governing Equation Groups

Combining the two balance laws, two constitutive laws, and two kinematic relations gives a strongly nonlinear and strongly coupled PDE–algebraic system.

Group (1): dynamics

$$\begin{cases} \rho A \dfrac{\partial^2 \boldsymbol{r}}{\partial t^2} - \dfrac{\partial \boldsymbol{n}}{\partial s} = \boldsymbol{f}_{\text{mag}}(\boldsymbol{B}) + \boldsymbol{f}_{\text{hydro}}\left(\dfrac{\partial \boldsymbol{r}}{\partial t}, \dfrac{\partial^2 \boldsymbol{r}}{\partial t^2}\right) + \boldsymbol{f}_{\text{cont}}(\boldsymbol{r}) + \boldsymbol{f}_{\text{gravity}}(\boldsymbol{r}), \\ J \dfrac{\partial \boldsymbol{w}}{\partial t} + \boldsymbol{w} \times (J\boldsymbol{w}) - \dfrac{\partial \boldsymbol{m}}{\partial s} - \dfrac{\partial \boldsymbol{r}}{\partial s} \times \boldsymbol{n} = \boldsymbol{l}_{\text{mag}}(\boldsymbol{B}) + \boldsymbol{l}_{\text{cont}}(\boldsymbol{r}). \end{cases} \tag{1}$$

Group (2): constitutive laws

$$\boldsymbol{m} = K_b(\boldsymbol{u} - \boldsymbol{u}_0), \qquad \boldsymbol{n} = K_s(\boldsymbol{\Gamma} - \boldsymbol{\Gamma}_0). \tag{2}$$

Group (3): geometric kinematics

$$\frac{\partial \boldsymbol{r}}{\partial s} = (1 + \varepsilon)\boldsymbol{d}_3, \qquad \frac{\partial \boldsymbol{d}_i}{\partial s} = \boldsymbol{u} \times \boldsymbol{d}_i. \tag{3}$$

1.5 PDE Reduction

The complete coupled system contains many intermediate variables. The aim is to eliminate these by writing the internal force directly in terms of derivatives of $\boldsymbol{r}(s, t)$.

1.5.1 Internal-force decomposition

The internal force is decomposed as

$$\boldsymbol{n} = EA(\nu - 1)\boldsymbol{t} + \nabla_s \cdot \boldsymbol{m} + K_s^{(r)}.$$

The three terms correspond, respectively, to axial stretching, bending-moment contribution, and shear force. Under the small-shear assumption the third term is neglected, giving

$$\boldsymbol{n} \approx EA(\nu - 1)\boldsymbol{t} + \nabla_s \cdot \boldsymbol{m}.$$

1.5.2 Expanded internal force

Using $v = \|\partial_s \boldsymbol{r}\|$ and the constitutive law for the bending moment, one obtains

$$\boldsymbol{n} \approx EA\left(\left|\frac{\partial \boldsymbol{r}}{\partial s}\right| - 1\right)\left|\frac{\partial \boldsymbol{r}}{\partial s}\right|^{-1}\frac{\partial \boldsymbol{r}}{\partial s} - \frac{\partial}{\partial s}\left(EI\frac{\partial^2 \boldsymbol{r}}{\partial s^2}\right).$$

This motivates the definition of the tangential nonlinear stiffness

$$k_{\tan}(s,t,r) := EA\left(\left|\frac{\partial \boldsymbol{r}}{\partial s}\right| - 1\right)\left|\frac{\partial \boldsymbol{r}}{\partial s}\right|^{-1} = EA\left(1 - \frac{1}{v}\right),$$

so that

$$\boldsymbol{n} = k_{\tan}\frac{\partial \boldsymbol{r}}{\partial s} - \frac{\partial}{\partial s}\left(EI\frac{\partial^2 \boldsymbol{r}}{\partial s^2}\right).$$

1.5.3 Gradient of the internal force

Differentiating with respect to s gives

$$\frac{\partial \boldsymbol{n}}{\partial s} = \frac{\partial}{\partial s}\left[k_{\tan}(s,t,r)\frac{\partial \boldsymbol{r}}{\partial s}\right] - \frac{\partial^2}{\partial s^2}\left(EI\frac{\partial^2 \boldsymbol{r}}{\partial s^2}\right). \tag{8}$$

The first term is a second-order axial restoring term with variable nonlinear stiffness. The second is a fourth-order bending-restoring term analogous to the Euler–Bernoulli beam operator.

1.5.4 Substitution into balance

Substituting the hydrodynamic force and the internal-force gradient into the translational equation, then collecting like terms, yields a standard inertial-plus-damping structure. First, the rod inertia and fluid added mass combine into

$$(\rho A + \rho_f A C_m)\mathbf{I}\,\frac{\partial^2 \boldsymbol{r}}{\partial t^2}.$$

Then the drag term is split so that the part proportional to $\partial_t \boldsymbol{r}$ is moved to the left-hand side as an effective damping term.

1.5.5 Final wave equation

The resulting PDE is

$$M(s,t)\frac{\partial^2 \boldsymbol{r}}{\partial t^2} + C\left(s,t,\frac{\partial \boldsymbol{r}}{\partial t}\right)\frac{\partial \boldsymbol{r}}{\partial t} = \frac{\partial}{\partial s}\left[k_{\tan}(s,t,r)\frac{\partial \boldsymbol{r}}{\partial s}\right] - \frac{\partial^2}{\partial s^2}\left(EI\frac{\partial^2 \boldsymbol{r}}{\partial s^2}\right) + \boldsymbol{F}. \tag{9}$$

with

$$M(s,t) := \rho A + \rho_f A C_m\,\mathbf{I}, \qquad C\left(s,t,\frac{\partial \boldsymbol{r}}{\partial t}\right) := \frac{1}{2}\rho_f D C_d\left|\frac{\partial \boldsymbol{r}}{\partial t} - \boldsymbol{V}_f\right|\mathbf{I},$$

and

$$\boldsymbol{F} := \rho_f A C_m \dot{\boldsymbol{V}}_f + \boldsymbol{f}_{\text{cont}}(\boldsymbol{r}) + \boldsymbol{f}_{\text{gravity}}(\boldsymbol{r}) + \boldsymbol{f}_{\text{mag}}(\boldsymbol{B}).$$

The structure is transparent: $M\ddot{\boldsymbol{r}}$ is inertia, $C\dot{\boldsymbol{r}}$ is nonlinear damping, the two spatial operators provide axial and bending restoring forces, and $\boldsymbol{F}$ is the driving term.

1.6 Transition to Discretization

Equation (9) is posed on a continuous domain $s \in [0, L]$ and therefore has infinitely many degrees of freedom. The next step is to approximate the field with finitely many basis functions so that the model becomes a computable system of ODEs.

1.7 Notation and Consistency

Several technical remarks should be recorded explicitly because they affect notation, dimensional consistency, and later derivations:

1. The strain vector in equation (iii) should be denoted by $\mathbf{\Gamma}$ rather than the position-vector symbol.
2. The third diagonal component of K_b is physically expected to be a torsional stiffness such as GJ.
3. The hydrodynamic drag term should include the outer diameter D for dimensional consistency.
4. The fluid-driving term in $\boldsymbol{F}$ should involve $\dot{\boldsymbol{V}}_f$ rather than $\boldsymbol{V}_f$ itself.
5. In the final PDE a typographical derivative $\partial\gamma/\partial s$ should be read as $\partial\boldsymbol{r}/\partial s$.

2. Galerkin Discretization

Part 1 produced a variable-coefficient nonlinear PDE on a continuous domain. The purpose of this part is to replace that infinite-dimensional model by a finite-dimensional ODE system via a Galerkin approximation.

2.1 Basic Galerkin idea

The approximate displacement field is written as

$$\hat{\boldsymbol{r}}(s,t) = N(s)\,\bar{q}_k(t), \qquad \text{(Approximation ansatz)}$$

where $N(s) = [N_1(s), \ldots, N_n(s)]^T$ is the vector of spatial basis/shape functions and $\bar{q}_k(t) \in \mathbb{R}^n$ is the vector of time-dependent generalized coordinates. Substituting the ansatz into the PDE introduces a residual $\bar{R}(s,t)$. Galerkin's condition imposes weighted orthogonality of the residual to the approximation space:

$$\int_0^L N^T(s)\,\bar{R}(s,t)\,ds \equiv 0. \tag{2.1}$$

This produces exactly n equations for the n unknown functions in $\bar{q}_k(t)$.

2.2 Residual form of the PDE

Substituting equation (9) from Part 1 yields

$$\int_0^L N^T(s)\left[M\ddot{\hat{\boldsymbol{r}}} + C\dot{\hat{\boldsymbol{r}}} - \frac{\partial}{\partial s}\left(k_{\tan}\frac{\partial \hat{\boldsymbol{r}}}{\partial s}\right) + \frac{\partial^2}{\partial s^2}\left(EI\frac{\partial^2 \hat{\boldsymbol{r}}}{\partial s^2}\right) - \bar{F}\right] ds = 0. \tag{10}$$

The low-order inertial and damping terms are projected directly. The stretching and bending terms involve spatial derivatives and are therefore treated by integration by parts.

2.3 Termwise Projection

2.3.1 Inertia term

Using $\ddot{\hat{\boldsymbol{r}}} = N(s)\ddot{\bar{q}}_k(t)$,

$$\int_0^L N^T M\ddot{\hat{\boldsymbol{r}}}\,ds = \left(\int_0^L N^T MN\,ds\right)\ddot{\bar{q}}_k(t).$$

2.3.2 Damping term

Likewise,

$$\int_0^L N^T C\hat{\dot{\boldsymbol{r}}}\, ds = \left(\int_0^L N^T C N\, ds\right) \dot{\bar{q}}_k(t).$$

2.3.3 Tangential stiffness term

The stretching contribution

$$\int_0^L N^T \frac{\partial}{\partial s}\left(k_{\tan}\frac{\partial \hat{\boldsymbol{r}}}{\partial s}\right) ds$$

is integrated by parts once. This transfers one derivative from the unknown field to the known test function:

$$N^T k_{\tan}\frac{\partial \hat{\boldsymbol{r}}}{\partial s}\Big|_0^L - \int_0^L \frac{dN^T}{ds}\, k_{\tan}\, \frac{dN}{ds}\, ds\; \bar{q}_k(t).$$

The boundary term is handled by the chosen boundary conditions.

2.3.4 Bending stiffness term

The bending contribution contains a fourth-order derivative and is integrated by parts twice:

$$\int_0^L N^T \frac{\partial^2}{\partial s^2}\left(EI\frac{\partial^2 \hat{\boldsymbol{r}}}{\partial s^2}\right) ds = \text{boundary terms} + \left(\int_0^L \frac{d^2N^T}{ds^2} EI \frac{d^2N}{ds^2}\, ds\right) \bar{q}_k(t).$$

The two natural boundary terms correspond, respectively, to end shear force and end bending moment.

2.4 Global Matrices and Force Vector

The extracted bilinear forms define the global matrices:

$$M_{\text{global}} := \int_0^L N^T M N\, ds, \tag{M}$$

$$C_{\text{global}} := \int_0^L N^T C N\, ds, \tag{C}$$

$$K_{\text{ten}} := \int_0^L \frac{dN^T}{ds}\, k_{\tan}\, \frac{dN}{ds}\, ds, \tag{K1}$$

$$K_{\text{bent}} := \int_0^L \frac{d^2N^T}{ds^2} EI \frac{d^2N}{ds^2}\, ds, \tag{K2}$$

$$\boldsymbol{F}_{\text{global}} := \int_0^L N^T \boldsymbol{F}\, ds. \tag{F}$$

The mass matrix is symmetric positive definite. The damping matrix has the same algebraic structure as the mass matrix. The two stiffness matrices encode axial and bending elasticity.

2.5 Finite-Dimensional ODE

After boundary terms are treated by the boundary conditions, the PDE becomes

$$M_{\text{global}}\ddot{\bar{q}}_k(t) + C_{\text{global}}\dot{\bar{q}}_k(t) + \big(K_{\text{ten}} + K_{\text{bent}}\big)\bar{q}_k(t) = \boldsymbol{F}_{\text{global}}. \tag{11}$$

In compact notation,

$$M\ddot{q} + C\dot{q} + Kq = F.$$

This is the central bridge from continuum mechanics to numerical computation.

2.6 Nonlinear Matrix Dependence

The compact linear-looking form above is exact only under small-deformation and linear-material assumptions. In the geometrically nonlinear setting the matrices depend on the current configuration.

2.6.1 $K(q)$ dependence

The axial stiffness depends on the local stretch through $k_{\tan}(q)$, which is an instance of stress stiffening. The bending contribution also becomes nonlinear because the exact curvature-displacement relation is nonlinear under large deformation.

2.6.2 $M(q)$ dependence

The kinetic energy is

$$T = \frac{1}{2}\int_0^L \rho A\,|\dot{r}|^2\,ds,$$

with $r = r(s,q)$. Under large deformation one has $\dot{r} = (\partial r/\partial q)\dot{q}$, which yields

$$M(q) = \int_0^L \rho A \left(\frac{\partial r}{\partial q}\right)^T \frac{\partial r}{\partial q}\,ds.$$

Thus the mass matrix inherits the geometry dependence of the position-to-coordinate map.

2.6.3 $C(q,\dot{q})$ and $F(q)$

Fluid damping may itself depend on speed, while geometry changes the projection of damping forces. External forces such as magnetic loads can also change direction and magnitude with the configuration.

2.6.4 Two levels of nonlinearity

It is useful to distinguish two levels of nonlinearity:

(a) **Geometric nonlinearity:** the shape functions remain fixed, but parameters inside the integrals depend on q.

(b) **Large deformation with configuration updates:** the shape functions themselves are updated with the configuration, $N(s) \mapsto N(s,q)$.

In both cases the finite-dimensional system is written abstractly as

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + K(q)q = F(q). \tag{2.2}$$

3. Hamiltonian Reformulation

Part 2 yielded a second-order ODE system. This part rewrites it as a first-order Hamiltonian-type system, makes the energy structure explicit, and prepares the ground for symplectic or structure-preserving algorithms.

3.1 Coordinates and Momenta

The generalized coordinate is $q \in \mathbb{R}^n$. The generalized momentum is defined via the Legendre transform:

$$p = \frac{\partial L}{\partial \dot{q}} = M(q)\dot{q}, \qquad \dot{q} = M^{-1}(q)p. \qquad \text{(Momentum definition)}$$

Because $M(q)$ is symmetric positive definite, its inverse exists.

3.2 Hamiltonian as total energy

The Hamiltonian is the sum of kinetic and potential energy,

$$H(q,p) = T(q,p) + V(q). \tag{3.1}$$

The kinetic energy is obtained by substituting $\dot{q} = M^{-1}(q)p$ into $T = \frac{1}{2}\dot{q}^T M(q)\dot{q}$:

$$T(q,p) = \frac{1}{2}p^T M^{-1}(q)p. \tag{3.2}$$

The inverse mass matrix plays the role of a Riemannian metric on configuration space. The potential energy is decomposed into stretching and bending parts,

$$V(q) = U_{\text{stretch}}(q) + U_{\text{bend}}(q),$$

and satisfies the gradient relation

$$\nabla_q V = K(q)q.$$

Because $K(q)$ is configuration dependent, $V(q)$ is not in general the quadratic form $\frac{1}{2}q^T Kq$.

3.3 Nonconservative Canonical Equations

The Hamiltonian system is written as

$$\begin{cases} \dot{q} = \dfrac{\partial H}{\partial p}, \\ \dot{p} = -\dfrac{\partial H}{\partial q} + Q(q, \dot{q}). \end{cases} \qquad \text{(Canonical form)}$$

The first equation reduces immediately to $\dot{q} = M^{-1}(q)p$. For the second equation,

$$\frac{\partial H}{\partial q} = \frac{1}{2}\nabla_q \left(p^T M^{-1}(q) p\right) + K(q)q,$$

so that

$$\dot{p} = -\frac{1}{2}\nabla_q \left(p^T M^{-1}(q) p\right) - K(q)q + Q(q, \dot{q}). \tag{3.3}$$

The three contributions have a clear interpretation:

1. a geometric inertial force from the q-dependence of the mass matrix,
2. a conservative restoring force from the potential-energy gradient,
3. a nonconservative generalized force Q.

3.4 Nonconservative Force Decomposition

The forcing term is decomposed as

$$Q(q, \dot{q}) = F_{\text{mag}} + F_{\text{cont}} - C(q, \dot{q})\dot{q}. \tag{3.4}$$

The last term is the dissipative damping force. If one wants a formulation purely in (q, p), one substitutes $\dot{q} = M^{-1}(q)p$.

3.5 Complete Hamiltonian system

Substituting the decomposition of Q gives

$$\begin{cases} \dot{q} = M^{-1}(q)p, \\ \dot{p} = -\nabla_q H + F_{\text{mag}} + F_{\text{cont}} - C(q, \dot{q})\nabla_p H. \end{cases} \tag{3.5}$$

Because $\nabla_p H = \dot{q}$, the damping contribution is exactly $-C\dot{q}$.

3.6 Energy evolution

Taking the total derivative of $H(q, p)$ along the motion yields

$$\frac{dH}{dt} = \frac{\partial H}{\partial q}\dot{q} + \frac{\partial H}{\partial p}\dot{p}. \tag{3.6}$$

Substituting the canonical equations and noting that the conservative terms cancel gives

$$\frac{dH}{dt} = \dot{q}^T \left(F_{\text{ext}} - C\dot{q}\right), \qquad F_{\text{ext}} = F_{\text{mag}} + F_{\text{cont}}. \tag{12}$$

This is the key energy identity of Part 3. It states that the Hamiltonian changes only through external work and damping dissipation:

- if $F_{\text{ext}} = 0$ and $C = 0$, the energy is conserved;
- if $F_{\text{ext}} = 0$ and $C > 0$, the energy decreases monotonically;
- if $F_{\text{ext}} \neq 0$ and $C > 0$, energy input and dissipation compete.

3.7 Hamilton-to-Birkhoff Transition

The damping term makes the system non-Hamiltonian in the strict symplectic sense because it cannot be represented by the standard constant skew-symmetric matrix J. This is the reason Part 5 introduces the Birkhoff framework: the fixed matrix J is replaced by a time-dependent skew-symmetric tensor $\mathcal{K}(z, t)$ so that structure-preserving geometry can be extended to nonconservative systems.

4. CHNN Framework

Part 4 develops the complex-coordinate formalism used to formulate, analyze, and train the CHNN. The original document mixes physical motivation, geometric structure, and algorithmic design. The English version below preserves that structure while translating it into standard mathematical language.

4.1 Magnetic-Dipole Force

The magnetic force is written as

$$\boldsymbol{F}_{\text{mag}} = (\boldsymbol{m} \cdot \nabla)\boldsymbol{B}(s,t). \tag{4.1}$$

Starting from the potential energy $U = -\boldsymbol{m} \cdot \boldsymbol{B}$, the force is obtained as

$$\boldsymbol{F} = -\nabla U = \nabla(\boldsymbol{m} \cdot \boldsymbol{B}).$$

Under the assumptions that $\boldsymbol{m}$ is constant in space and the magnetic field is magnetostatic ($\nabla \times \boldsymbol{B} = 0$), one recovers

$$\boldsymbol{F} = \nabla(\boldsymbol{m} \cdot \boldsymbol{B}) = (\boldsymbol{m} \cdot \nabla)\boldsymbol{B}. \tag{4.2}$$

Three immediate physical observations follow: the force vanishes in a spatially uniform field, its direction depends on the coupling between $\boldsymbol{m}$ and the gradient of $\boldsymbol{B}$, and the operator $\boldsymbol{m} \cdot \nabla$ measures directional variation along $\boldsymbol{m}$.

4.2 Poisson Brackets

The Poisson bracket is defined by

$$\{f, g\} = \sum_{i=1}^{N} \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right). \tag{14}$$

The time evolution of an observable is

$$\frac{df}{dt} = \{f, H\} + \frac{\partial f}{\partial t}.$$

The usual algebraic properties are recorded explicitly: antisymmetry, linearity, the Leibniz rule, and the Jacobi identity. The canonical commutation relations are

$$\{q_i, q_j\} = 0, \qquad \{p_i, p_j\} = 0, \qquad \{q_i, p_j\} = \delta_{ij},$$

which imply $\{q_i, H\} = \partial H/\partial p_i$ and $\{p_i, H\} = -\partial H/\partial q_i$.

4.3 Metriplectic Extension

In matrix form the Hamiltonian equations are

$$\dot{\mathbf{z}} = J\nabla H, \qquad J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}.$$

To model dissipation, introduce a symmetric positive-semidefinite dissipation matrix $D(\mathbf{z})$ and write

$$\dot{\mathbf{z}} = (J - D)\nabla H + \boldsymbol{F}_{\text{ext}}. \tag{15}$$

The energy balance becomes

$$\frac{dH}{dt} = (\nabla H)^T J \nabla H - (\nabla H)^T D \nabla H + (\nabla H)^T \boldsymbol{F}_{\text{ext}}. \tag{26}$$

Since the skew-symmetry of J implies $(\nabla H)^T J \nabla H = 0$, the conservative part drops out and only dissipation and external work remain. The manuscript also records the standard metriplectic evolution law

$$\frac{df}{dt} = \{f, \mathcal{H}\} + \mathcal{T}(f, \mathcal{S}),$$

which combines a Poisson bracket with a symmetric dissipative bracket.

4.4 Complex Hamiltonian

The complex-valued generating function is introduced as

$$\mathcal{H}(q,p) = H(q,p) + i\big(G_1(q,p) + G_2(q,p)\big) = H(q,p) + iG(q,p). \tag{16}$$

The real part encodes the conservative Hamiltonian, while the imaginary part collects dissipative and auxiliary contributions. This is the key algebraic move that allows the subsequent complex-coordinate formulation of the dynamics.

4.5 Complex Coordinates

The complex canonical coordinates are defined by

$$\xi = \frac{1}{\sqrt{2}}(q + ip), \qquad \bar{\xi} = \frac{1}{\sqrt{2}}(q - ip). \tag{17}$$

The associated Wirtinger derivatives are

$$\frac{\partial}{\partial \xi} = \frac{1}{2}\left(\frac{\partial}{\partial q} - i\frac{\partial}{\partial p}\right), \qquad \frac{\partial}{\partial \bar{\xi}} = \frac{1}{2}\left(\frac{\partial}{\partial q} + i\frac{\partial}{\partial p}\right). \tag{18}$$

In these coordinates the symplectic two-form becomes

$$\omega = dq \wedge dp = i\, d\xi \wedge d\bar{\xi}. \tag{4.3}$$

The complex-coordinate Poisson bracket gives $\{\xi, \bar{\xi}\} = -i$, which is the direct analogue of the canonical real bracket.

4.6 Complex Equations of Motion

The complex velocity is

$$\dot{\xi} = \frac{1}{\sqrt{2}}(\dot{q} + i\dot{p}).$$

After substituting the Hamiltonian equations and separating conservative and dissipative contributions, one obtains

$$\dot{\xi} = -\sqrt{2}\, i\, \frac{\partial H}{\partial \bar{\xi}} + P(\xi, \bar{\xi}), \tag{21}$$

where $P(\xi, \bar{\xi})$ encodes nonconservative contributions. The complex Poisson bracket is defined as

$$\{f, g\}_{\mathfrak{C}} = -i\left(\frac{\partial f}{\partial \xi}\frac{\partial g}{\partial \bar{\xi}} - \frac{\partial f}{\partial \bar{\xi}}\frac{\partial g}{\partial \xi}\right), \tag{4.4}$$

so that the equation of motion can be written compactly as

$$\dot{\xi} = \sqrt{2}\{\xi, H\}_{\mathfrak{C}} + R(\xi, \bar{\xi}). \tag{24}$$

Finally, by absorbing the dissipative correction into a complex-valued effective Hamiltonian,

$$\tilde{H} = \sqrt{2}\big(H + i\tilde{P}\big), \tag{4.5}$$

one obtains the unified form

$$\dot{\xi} = -i\frac{\partial \tilde{H}}{\partial \bar{\xi}}. \tag{25}$$

This equation is the formal target modeled by the CHNN.

4.7 Action–Angle Variables

For clarity, this subsection first considers a single modal reference oscillator. In this scalar setting, $q, p \in \mathbb{R}$, and $M > 0$, $K > 0$ denote the modal mass and modal stiffness. The corresponding natural frequency is

$$\omega_0 = \sqrt{\frac{K}{M}}.$$

For this one-degree-of-freedom oscillator, the normalized complex coordinate is introduced as

$$\xi = \frac{1}{\sqrt{2}}\left(\sqrt{M\omega_0}\, q + \frac{ip}{\sqrt{M\omega_0}}\right),$$

and one may write

$$\xi = \sqrt{I}\, e^{i\theta}, \qquad \bar{\xi} = \sqrt{I}\, e^{-i\theta}. \tag{22}$$

Here I is the action variable and θ is the angle variable. The symplectic form becomes $\omega = dI \wedge d\theta$, so (I, θ) form an action–angle pair. In this representation, the Cartesian phase-space variables (q, p) are reorganized into an amplitude-like invariant I and a phase variable θ. For the conservative reference

oscillator, I remains constant and θ evolves almost linearly in time.

For the full Galerkin-discretized rod model, the situation is different. In general, $q, p \in \mathbb{R}^n$ and $M, K \in \mathbb{R}^{n\times n}$, so the Hamiltonian phase space has dimension $2n$. Therefore, the scalar expression $\omega_0 = \sqrt{K/M}$ should not be interpreted as a direct matrix division. A multi-degree-of-freedom action–angle formulation requires modal diagonalization, for example through the generalized eigenvalue problem

$$K\phi_i = \omega_i^2 M\phi_i.$$

After mass-normalized modal transformation, each decoupled mode can be assigned its own pair of action–angle variables (I_i, θ_i). Thus, the formula above should be understood as the one-mode prototype used to explain the geometric meaning of action–angle coordinates.

The perturbed modal dynamics may then be written schematically as

$$\dot{I}_i = \varepsilon f_i(I, \theta), \qquad \dot{\theta}_i = \omega_i(I) + \varepsilon g_i(I, \theta),$$

where weak perturbations slowly modulate the actions while the angles carry the fast oscillatory motion. This viewpoint is useful for discussing invariant-torus persistence and long-time structural fidelity.

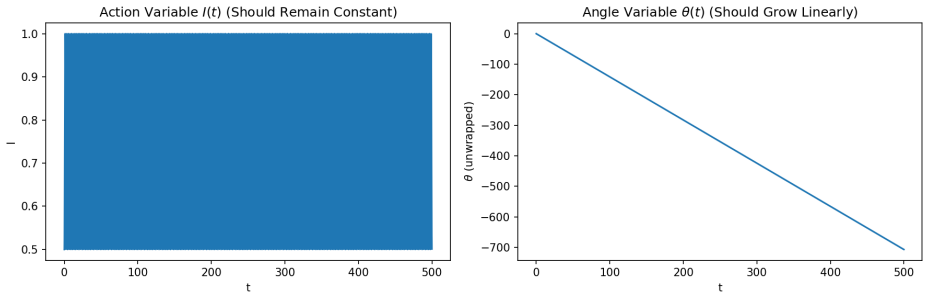


Figure 4.1: Action–angle representation of a reference oscillator. The figure illustrates how the phase-space trajectory is rewritten in terms of the action variable I and the angle variable θ, which separates slow amplitude modulation from fast phase rotation.

4.8 KAM Stability Viewpoint

The manuscript invokes the Diophantine nonresonance condition

$$|\langle k, \omega(I)\rangle| \geq \frac{\gamma}{|k|^\tau}, \qquad k \in \mathbb{Z}^N \setminus \{0\},\ \gamma > 0,\ \tau > N - 1,$$

as the standard hypothesis under which invariant tori persist under sufficiently small perturbations. The cited KAM viewpoint is used as a structural motivation: preserving the geometric structure of the flow is expected to improve long-time qualitative fidelity.

4.9 Complex Analyticity

Writing a complex-valued function as $f(x+iy) = u(x, y) + iv(x, y)$, the Cauchy–Riemann equations are

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}. \tag{27}$$

A compact equivalent representation is

$$\frac{\partial f}{\partial x} = -i\frac{\partial f}{\partial y}. \tag{28}$$

The document also notes basic properties of analytic functions and recalls Liouville's theorem as a restriction on bounded entire activation functions in a complex network.

Figure 4.2 summarizes the activation functions considered in the complex-network component. The point of the comparison is not only numerical expressivity, but also compatibility with the analytic or approximately analytic structure imposed by the complex Hamiltonian representation. Bounded entire activations are theoretically restrictive, while practical complex networks typically trade exact analyticity for trainability and stable gradients.

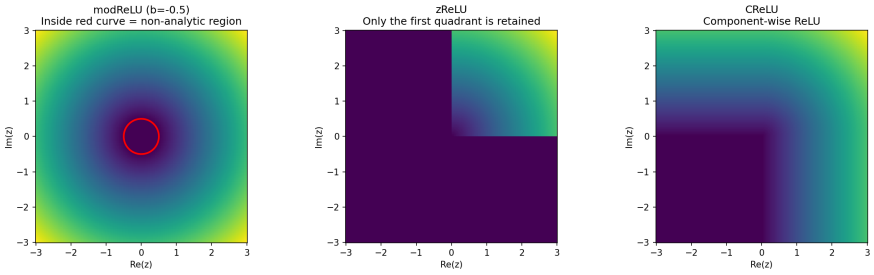


Figure 4.2: Activation functions used in the complex-valued network discussion. The comparison highlights the practical tension between complex differentiability, nonlinear expressivity, and numerical stability.

4.10 Birkhoff Equation

The Birkhoff equation is written as

$$\left(\frac{\partial F_j}{\partial z_i} - \frac{\partial F_i}{\partial z_j}\right)\dot{z}_j - \left(\frac{\partial B}{\partial z_i} + \frac{\partial F_i}{\partial t}\right) = 0. \tag{$\star$}$$

The manuscript then explains how the damped equation from Part 3,

$$\dot{q} = M^{-1}(q)p, \qquad \dot{p} = -\frac{\partial H}{\partial q} + F_{\text{ext}} - CM^{-1}p, \tag{12}$$

may be embedded into the Birkhoff formalism.

4.11 Generating Functions

Auxiliary variables are introduced via

$$\hat{Q} = \hat{p} - p, \quad \hat{P} = \hat{q} - q, \quad Q = \tfrac{1}{2}(\hat{q} + q), \quad P = -\tfrac{1}{2}(\hat{p} + p), \tag{30}$$

and the associated linear transformation is represented by the Jacobian matrix

$$\alpha_* = \begin{pmatrix} 0 & I & 0 & -I \\ I & 0 & -I & 0 \\ \frac{1}{2}I & 0 & \frac{1}{2}I & 0 \\ 0 & -\frac{1}{2}I & 0 & -\frac{1}{2}I \end{pmatrix}. \tag{31}$$

The manuscript cites two results, Theorem (A) on the existence of the generating function and Theorem (B) on its Taylor-series expansion. These are used later to derive structure-preserving difference schemes.

4.12 Birkhoff Discretization

Let $w = (P, Q)$ and define a generating-function expansion

$$\phi_w(w, t, t_0) = \sum_{k=0}^{\infty} (t - t_0)^k \phi_w^{(k)}(w, t_0).$$

The first coefficient is

$$\phi_w^{(1)}(w, t_0) = \begin{pmatrix} -\frac{1}{2} P^T \dfrac{\partial M^{-1}}{\partial Q} P - K(Q)Q + F_{\text{ext}} + CM^{-1}P \\ -M^{-1}(Q)P \end{pmatrix}, \tag{4.6}$$

and the higher-order coefficients satisfy the recursive rule

$$\phi_w^{(k)}(w, t_0) = \frac{1}{k} D_t \phi_w^{(k-1)}(w, t_0), \qquad k \geq 2.$$

Truncation at first order gives the discrete update equations (33)–(34). In the constant-matrix case this leads to the closed-form scheme

$$\begin{cases} p_{k+1} = p_k \dfrac{1 - \varepsilon}{1 + \varepsilon} - K q_k \dfrac{\delta}{1 + \varepsilon} + F_{\text{ext}} \dfrac{\delta}{1 + \varepsilon}, \\ q_{k+1} = q_k + \dfrac{\delta}{2M}(p_{k+1} + p_k), \end{cases} \tag{36}$$

where the first-order parameter ε is determined by the damping and stiffness constants. The second-order correction $\phi_w^{(2)}$ can also be written explicitly, together with the general Kth-order truncation operator

$$\Gamma_K \phi_w(w, t; t_0) = \sum_{k=0}^{K} (t - t_0)^k \phi_w^{(k)}(w, t_0).$$

4.13 CHNN Training

The CHNN parameterizes an effective complex Hamiltonian

$$\tilde{H}_\lambda(\xi, \bar{\xi}) = \sqrt{2}\big(H_\lambda(\xi, \bar{\xi}) + i\tilde{P}_\lambda(\xi, \bar{\xi})\big), \tag{4.7}$$

and predicts the vector field through

$$\dot{\xi}_{\text{pred}} = -i \frac{\partial \tilde{H}_\lambda}{\partial \bar{\xi}}\bigg|_{\xi=\xi_t}. \tag{4.8}$$

The training objective is composed of five terms:

$$L_{\text{energy}} = \left\| \frac{\partial \tilde{H}_\lambda}{\partial \xi}\dot{\xi}_{\text{obs}} + \frac{\partial \tilde{H}_\lambda}{\partial \bar{\xi}}\dot{\bar{\xi}}_{\text{obs}} \right\|^2, \tag{4.9}$$

$$L_{\text{hamilton}} = \left\| \dot{\xi}_{\text{obs}} + i\frac{\partial \tilde{H}_\lambda}{\partial \bar{\xi}} \right\|^2, \tag{4.10}$$

$$L_{\text{trajectory}} = \left\| \xi^{\text{obs}}_{t+\Delta t} - \text{Integrate}(\xi_t, \dot{\xi}_{\text{pred}}, \Delta t) \right\|^2, \tag{4.11}$$

$$L_{\text{Jacobi}} = \left| \det \frac{\partial(\xi^{\text{pred}}_{t+\Delta t}, \bar{\xi}^{\text{pred}}_{t+\Delta t})}{\partial(\xi_t, \bar{\xi}_t)} - 1 \right|^2, \tag{4.12}$$

$$L_{\text{symplectic}} = \left\| \frac{\partial u_\lambda}{\partial x} - \frac{\partial v_\lambda}{\partial y} \right\|^2 + \left\| \frac{\partial u_\lambda}{\partial y} + \frac{\partial v_\lambda}{\partial x} \right\|^2. \tag{4.13}$$

The total loss is a weighted combination

$$\mathcal{L}_{\text{total}} = \alpha_1 L_{\text{energy}} + \alpha_2 L_{\text{hamilton}} + \alpha_3 L_{\text{trajectory}} + \alpha_4 L_{\text{Jacobi}} + \alpha_5 L_{\text{symplectic}}. \tag{4.14}$$

For dissipative systems, a practical weighting strategy is to emphasize the Hamiltonian-matching and trajectory terms.

Figure 4.3 reports the representative CHNN training behavior. The loss curve indicates whether the learned complex Hamiltonian vector field fits the observed dynamics, while the trajectory and derivative comparisons show how local vector-field accuracy translates into rollout behavior. The figure should therefore be read as a diagnostic of both instantaneous Hamiltonian matching and accumulated prediction error.

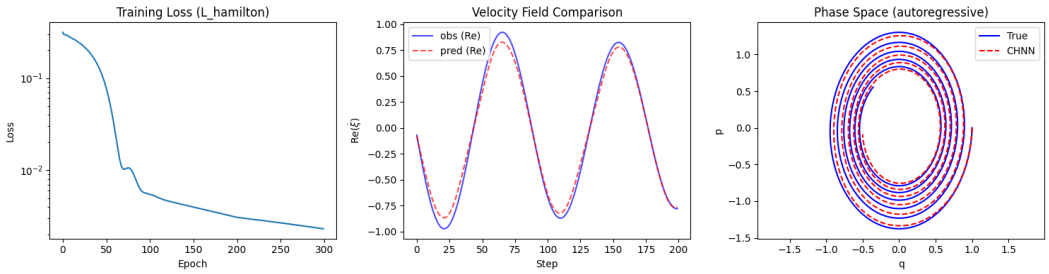


Figure 4.3: Representative CHNN training result. The panels compare training loss, learned vector-field components, and rollout behavior, thereby linking the Hamiltonian loss to trajectory-level prediction quality.

4.14 High-Order Scheme Construction

A symbolic recursion is proposed to build high-order updates. To avoid dimensional ambiguity, the first-order seed scheme is first written for a single modal oscillator. In this scalar prototype, $q_k, p_k \in \mathbb{R}$,

$M > 0$, and $K > 0$, so that M and K are modal parameters rather than full matrices. Define

$$\varepsilon_0 = \frac{K\delta^2}{4M}.$$

The first-order update is

$$p_{k+1} = p_k \frac{1-\varepsilon_0}{1+\varepsilon_0} - Kq_k \frac{\delta}{1+\varepsilon_0},$$

$$q_{k+1} = q_k + \frac{\delta}{2M}(p_{k+1} + p_k).$$

This scalar scheme serves as the seed of the high-order recursion. It should be interpreted as a one-mode structure-preserving prototype rather than the full discretization of the Galerkin rod system.

For the full multi-degree-of-freedom system, $q_k, p_k \in \mathbb{R}^n$ and $M, K \in \mathbb{R}^{n\times n}$; the scalar divisions in the above formulas must therefore be replaced by matrix operations or linear solves. A matrix-level midpoint form may be written as

$$p_{k+1} = p_k - \frac{\delta}{2}K(q_{k+1} + q_k),$$

$$q_{k+1} = q_k + \frac{\delta}{2}M^{-1}(p_{k+1} + p_k).$$

Equivalently, eliminating q_{k+1} gives

$$\left(I + \frac{\delta^2}{4}KM^{-1}\right)p_{k+1} = \left(I - \frac{\delta^2}{4}KM^{-1}\right)p_k - \delta K q_k.$$

Thus, in the multi-degree-of-freedom case, the update requires solving a linear system rather than using the scalar factor $(1+\varepsilon_0)^{-1}$. If damping or external forcing is included, these terms should enter through the nonconservative force term or the Birkhoff generating-function correction, rather than being hidden in the scalar stiffness parameter.

The general high-order construction is then obtained by symbolic differentiation of the generating-function coefficients and by using fixed-point iteration to solve the resulting implicit equations.

4.15 Numerical experiments

The numerical experiments are used to check whether the theoretical structures above produce the expected qualitative behavior under several physical regimes. Rather than treating the plots as detached supplementary material, the figures are interpreted here as part of the experimental analysis.

Figure 4.4 compares five representative dynamical scenarios. The conservative case provides the reference closed or nearly closed phase-space behavior. The damped cases show contraction toward lower-energy states, with stronger damping accelerating the decay. The forced and forced-damped cases demonstrate how external input competes with dissipation and can generate persistent response patterns. Taken together, the five panels verify that the numerical model distinguishes conservative evolution, dissipative decay, external excitation, and mixed forced-dissipative dynamics.

The convergence behavior of the structure-preserving integrator is summarized in Figure 4.5. The log–log slope is the key quantity: it indicates the empirical order of accuracy when the time step is refined. This figure is therefore used to verify that the implemented discretization behaves consistently with the expected order of the underlying Birkhoff/symplectic construction, rather than merely producing visually

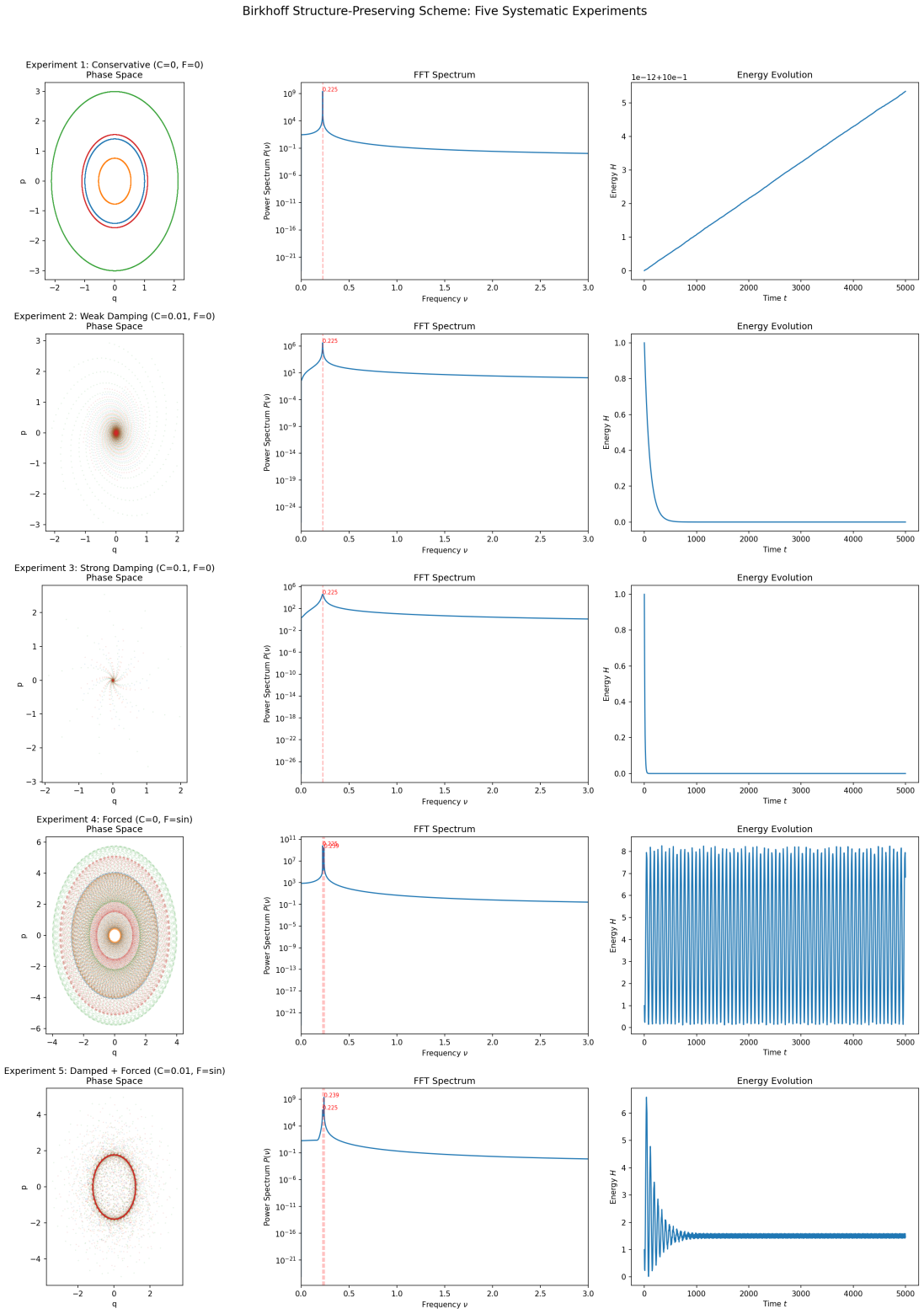


Figure 4.4: Five representative dynamical experiments. The comparison separates conservative, damped, forced, and forced-damped regimes and shows how phase-space geometry, temporal response, energy evolution, and spectral content change across physical conditions.

plausible trajectories.

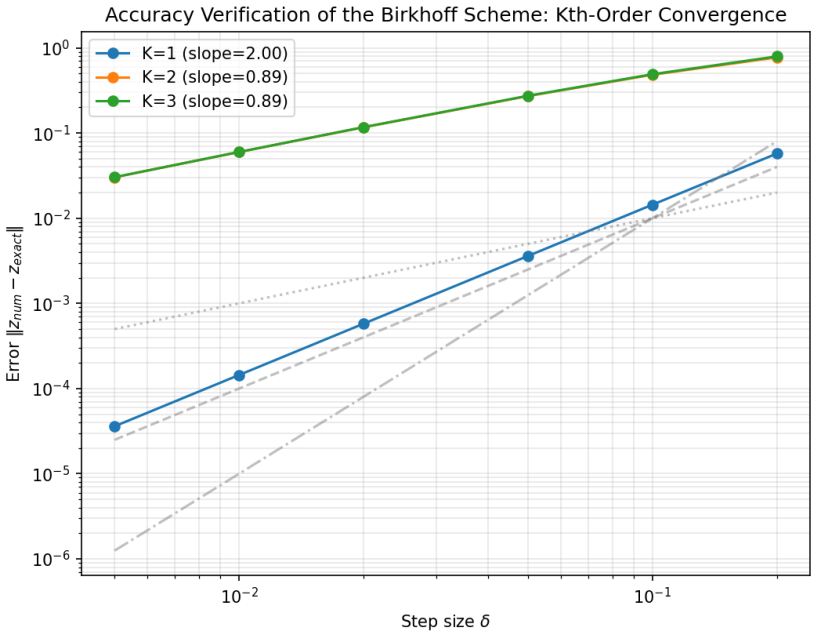


Figure 4.5: Log–log convergence test for the structure-preserving discretization. The slope of the error curve provides an empirical estimate of the method's convergence order under time-step refinement.

4.16 Control Compensation

Three compensation strategies are proposed:

Scheme A: external-force embedding (recommended),

Scheme B: additive correction,

Scheme C: correction at the generating-function level.

The manuscript compares them conceptually and numerically. It also proposes design principles for the control term, four typical control forms, and three evaluation metrics used in experiments: trajectory error, symplectic deviation, and energy drift.

Figure 4.6 compares the resulting compensation behavior. The purpose of this comparison is to identify whether the added control term reduces trajectory error without destroying the geometric structure that the Birkhoff formulation is designed to preserve. A satisfactory compensation scheme should therefore improve tracking while keeping symplectic deviation and artificial energy drift under control.

4.17 Normalization Factor

The factor $\sqrt{2}$ should be retained in equation (25), because it has already been absorbed into the definition of the effective Hamiltonian. This convention is then used consistently across the subsequent formula blocks.

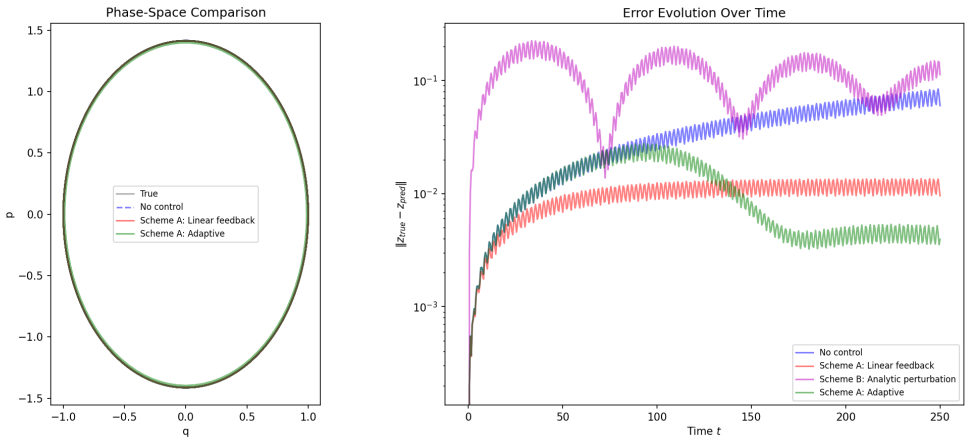


Figure 4.6: Comparison of control-compensation strategies for unmodeled dynamics. The figure is used to assess the trade-off between trajectory correction, structure preservation, and energy behavior.

Part 4 Logic

The logical progression of Part 4 can be summarized as follows: magnetic forcing motivates the external nonconservative input; Poisson geometry provides the conservative backbone; metriplectic and Birkhoff ideas extend the framework to dissipation; complex coordinates and Wirtinger calculus make the structure compact; generating functions produce structure-preserving discretizations; and the CHNN approximates the resulting complex Hamiltonian vector field while being trained with geometry-aware losses.

5. $\mathcal{K}$-Symplectic Algorithms

Part 5 develops the $\mathcal{K}$-symplectic viewpoint for nonconservative systems by following the treatment in Kang Feng and Mengzhao Qin. The goal is to extend structure-preserving numerical ideas from standard Hamiltonian systems to damped or otherwise nonconservative dynamics.

5.1 Birkhoff System Equations

The motivating model is the nonlinear damped second-order system

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + K(q)q = F_{\text{ext}}.$$

Because of the dissipative term $C\dot{q}$, such a system is not representable by the standard Hamiltonian form $\dot{z} = J\nabla H$ with constant skew-symmetric matrix J. The Birkhoff equation therefore introduces a more flexible geometric tensor:

$$\left(\frac{\partial F_j}{\partial z_i} - \frac{\partial F_i}{\partial z_j}\right)\frac{dz_j}{dt} = \frac{\partial B(z,t)}{\partial z_i} + \frac{\partial F_i(z,t)}{\partial t}, \qquad i = 1,\dots,2n. \tag{$\star$}$$

The Birkhoff tensor and the generalized force term are defined by

$$\mathcal{K}_{ij} = \frac{\partial F_j}{\partial z_i} - \frac{\partial F_i}{\partial z_j}, \qquad D_i(z,t) = -\frac{\partial B}{\partial z_i} - \frac{\partial F_i}{\partial t}. \tag{5.1}$$

With this notation one obtains the equivalent forms

$$\mathcal{K}_{ij}(z)\frac{dz_j}{dt} = \frac{\partial B(z)}{\partial z_i}, \tag{2.2}$$

$$\mathcal{K}_{ij}(z,t)\frac{dz_j}{dt} = \frac{\partial B}{\partial z_i} + \frac{\partial F_i}{\partial t}, \tag{2.3}$$

$$\mathcal{K}_{ij}(z,t)\frac{dz_j}{dt} + D_i(z,t) = 0. \tag{2.4}$$

In matrix notation,

$$\mathcal{K}(z,t)\frac{dz}{dt} = \nabla B + \frac{\partial F}{\partial t}. \tag{2.8}$$

5.2 Self-Adjointness

The self-adjointness conditions are

$$\begin{cases} \text{(I) } \mathcal{K}_{ij} + \mathcal{K}_{ji} = 0, \\ \text{(II) } \dfrac{\partial \mathcal{K}_{ij}}{\partial z_k} + \dfrac{\partial \mathcal{K}_{jk}}{\partial z_i} + \dfrac{\partial \mathcal{K}_{ki}}{\partial z_j} = 0, \\ \text{(III) } \dfrac{\partial D_i}{\partial z_j} - \dfrac{\partial D_j}{\partial z_i} = \dfrac{\partial \mathcal{K}_{ij}}{\partial t}. \end{cases} \tag{2.5}$$

The associated symplectic two-form is

$$\Omega = \sum_{i<j} \mathcal{K}_{ij}(z,t)\, dz_i \wedge dz_j. \tag{5.2}$$

Because

$$\Omega = d\left(\sum_i F_i\, dz_i \right),$$

it is closed: $d\Omega = 0$. The manuscript interprets this as the exact differential of a one-form, thereby generalizing standard symplectic geometry to the Birkhoff setting.

5.3 $\mathcal{K}$-Symplectic Mappings

A map $\hat{z} = g(z, t, t_0)$ is $\mathcal{K}$-symplectic if it satisfies

$$\left(\frac{\partial g}{\partial z}\right)^T \mathcal{K}(\hat{z}, \hat{t}) \left(\frac{\partial g}{\partial z}\right) = \mathcal{K}(z,t). \tag{5.3}$$

For a one-step method this becomes

$$\left(\frac{\partial g^k}{\partial z^k}\right)^T \mathcal{K}(z^{k+1}, t_{k+1}) \frac{\partial g^k}{\partial z^k} = \mathcal{K}(z^k, t_k).$$

An auxiliary coordinate transformation α is then introduced to map $(\hat{z}, z)$ to auxiliary variables $(\hat{w}, w)$. The associated invertibility condition on the Jacobian blocks of α is one of the technical prerequisites for the generating-function theory.

5.4 Taylor Recursion

The generating function satisfies an evolution equation of the form

$$\frac{\partial}{\partial t}\phi_w(w,t,t_0) = \mathcal{A}\big(\phi_w, w, \phi_{ww}, t, t_0\big). \tag{4.3}$$

Theorem (A) asserts the existence of a local generating function under the invertibility assumptions. Theorem (B) then expands it as

$$\phi_w(w,t,t_0) = \sum_{k=0}^{\infty} (t-t_0)^k \phi_w^{(k)}(w,t_0). \tag{4.9}$$

The coefficients satisfy a recursive rule beginning with

$$\phi_w^{(0)}(w,t_0) = f(w,t_0,t_0), \qquad \phi_w^{(1)}(w,t_0) = \mathcal{A}\big(\phi_w^{(0)}, w, \phi_{ww}^{(0)}, t_0, t_0\big), \tag{4.10}$$

and continuing via repeated total differentiation with respect to time.

5.5 $\mathcal{K}$-Symplectic Discretization

Truncating the Taylor series of the generating function and transforming back from the auxiliary variables to (q,p) yields a damped structure-preserving scheme. In the constant-matrix case the resulting update reads

$$q_{k+1} = q_k + \frac{1}{2} M^{-1}(p_{k+1} + p_k)\,\delta, \tag{†-a}$$

$$p_{k+1} = p_k \frac{1-\varepsilon}{1+\varepsilon} - K q_k \frac{\delta}{1+\varepsilon} + (\Delta + F_{\text{ext}}) \frac{\delta}{1+\varepsilon}. \tag{†-b}$$

The parameter ε acts as a damping factor and, in the pure damping case, its rational form

$$\frac{1-\varepsilon}{1+\varepsilon} = \frac{1 - C\delta/2}{1 + C\delta/2}$$

is identified in the manuscript as the $[1/1]$ Padé approximation to the exponential decay factor.

The preserved weighted symplectic structure is written as

$$e^{\int_0^{t_{k+1}} C\,ds}\, dq_{k+1} \wedge dp_{k+1} = e^{\int_0^{t_k} C\,ds}\, dq_k \wedge dp_k, \tag{5.4}$$

or, for constant damping rate ν,

$$e^{\nu t_{k+1}}\, dq_{k+1} \wedge dp_{k+1} = e^{\nu t_k}\, dq_k \wedge dp_k. \tag{5.5}$$

This is the damped analogue of exact symplectic preservation.

5.6 Error Control

Three complementary mechanisms explain why the structure-preserving discretization controls error growth.

1. The modified equation of the discrete scheme remains a Birkhoff system, so the geometric structure is inherited at the backward-error level.

2. The energy error stays bounded rather than drifting secularly. One may therefore write bounds of the form $|B(z_k) - B(z_0)| \le C_1 \delta^m$ over long intervals, in contrast to the linearly growing worst case $C_2 k \delta^m$ of non-structure-preserving methods.

3. The implicit midpoint-type structure is not accidental; it is forced by the symplectic condition itself.

A numerical test on the damped oscillator

$$\ddot{q} + \nu\dot{q} + q = 0, \qquad q(0) = 1, \quad p(0) = -1, \qquad t \in [0, 25],$$

illustrates trajectory and energy errors. The associated discussion connects the long-time behavior to the KAM-motivated view that geometric fidelity matters for qualitative accuracy.

5.7 Overall summary

The theoretical chain developed in this part is as follows. Starting from a nonconservative second-order system, one rewrites it as a Birkhoff system, constructs the associated two-form and self-adjointness conditions, introduces $\mathcal{K}$-symplectic maps and their generating functions, derives Taylor-recursive structure-preserving discretizations, and then explains why preserving the geometric structure suppresses spurious long-time drift. In concise form: *Birkhoff geometry is the natural extension of symplectic geometry to damped systems, and $\mathcal{K}$-symplectic algorithms inherit the resulting stability advantages.*